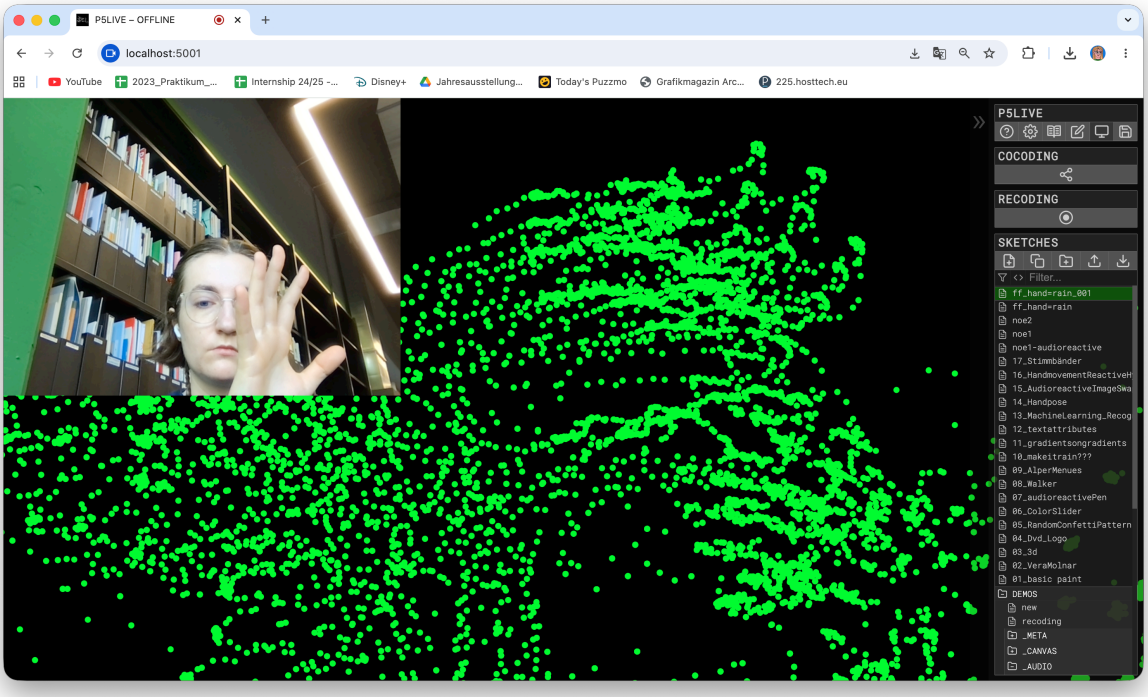
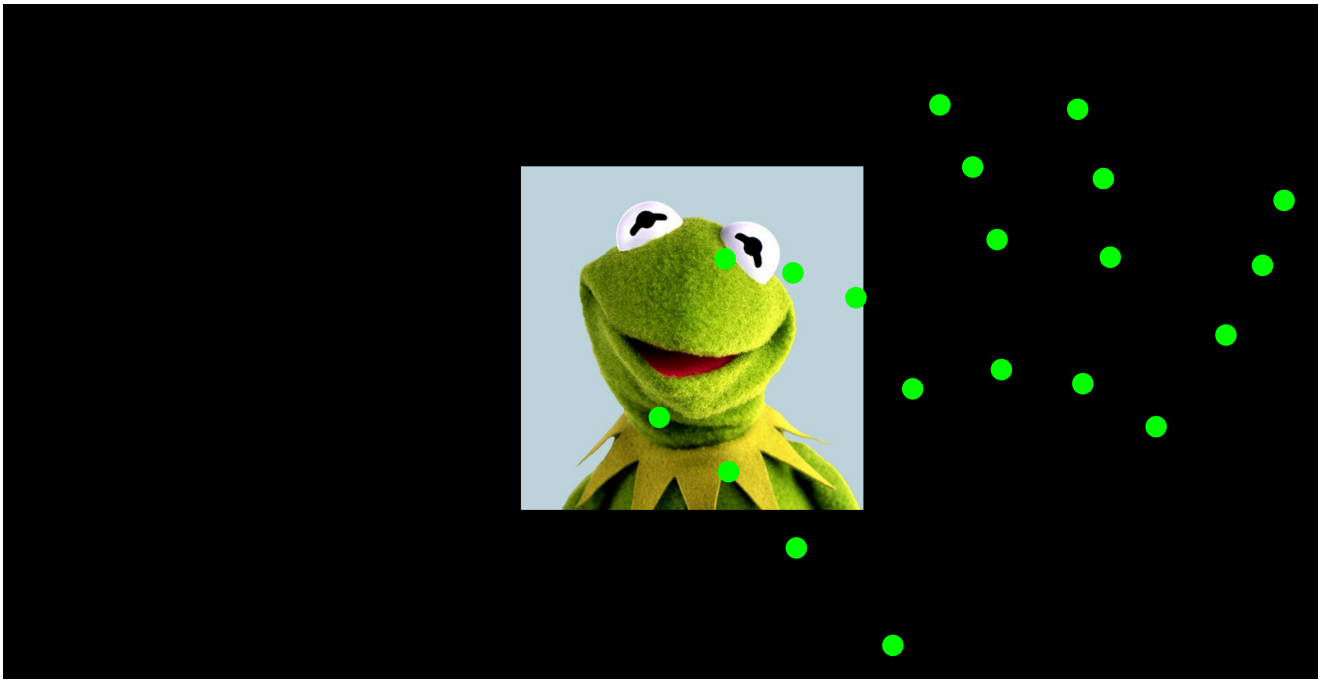
Handtracking



```
let libs = [
    'https://unpkg.com/ml5@1/dist/ml5.min.js',
    'https://unpkg.com/hydra-synth',
    'https://cdn.jsdelivr.net/gh/ffd8/hy5@main/hy5.js'
]

let handPose;
let video;
let hands = [];

function preload() {
  // Load the handPose model
  handPose = ml5.handPose();
}

function setup() {
  createCanvas(windowWidth, windowHeight);
     video = createCapture(VIDEO, { flipped: true });
  // Create the webcam video and hide it
  video = createCapture(VIDEO);
  video.size(windowWidth, windowHeight);
  video.hide();
  // start detecting hands from the webcam video
  handPose.detectStart(video, gotHands);
}

function draw() {
  // Draw the webcam video
  //image(video, 0, 0, width, height);

  // Draw all the tracked hand points
  for (let i = 0; i < hands.length; i++) {
    let hand = hands[i];
    for (let j = 0; j < hand.keypoints.length; j++) {
      let keypoint = hand.keypoints[j];
      fill(0, 255, 0);
      noStroke();
      circle(keypoint.x, keypoint.y, 10);
    }
  }
}

// Callback function for when handPose outputs data
function gotHands(results) {
  // save the output to the hands variable
  hands = results;
}
```

Handpuppet



```
let libs = [
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  'https://unpkg.com/hydra-synth',
  'https://cdn.jsdelivr.net/gh/ffd8/hy5@main/hy5.js'
]

let handPose;
let video;
let hands = [];
let kermit;

function preload() {
  handPose = ml5.handPose();
  kermit = loadImage(
    'https://lumiere-
a.akamaihd.net/v1/images/character_themuppets_kermit_b77a431b.jpeg');
}

function setup() {
  createCanvas(windowWidth, windowHeight);

  video = createCapture(VIDEO);
  video.size(windowWidth, windowHeight);
  video.hide();

  handPose.detectStart(video, gotHands);
}

function draw() {
  background(0);
```
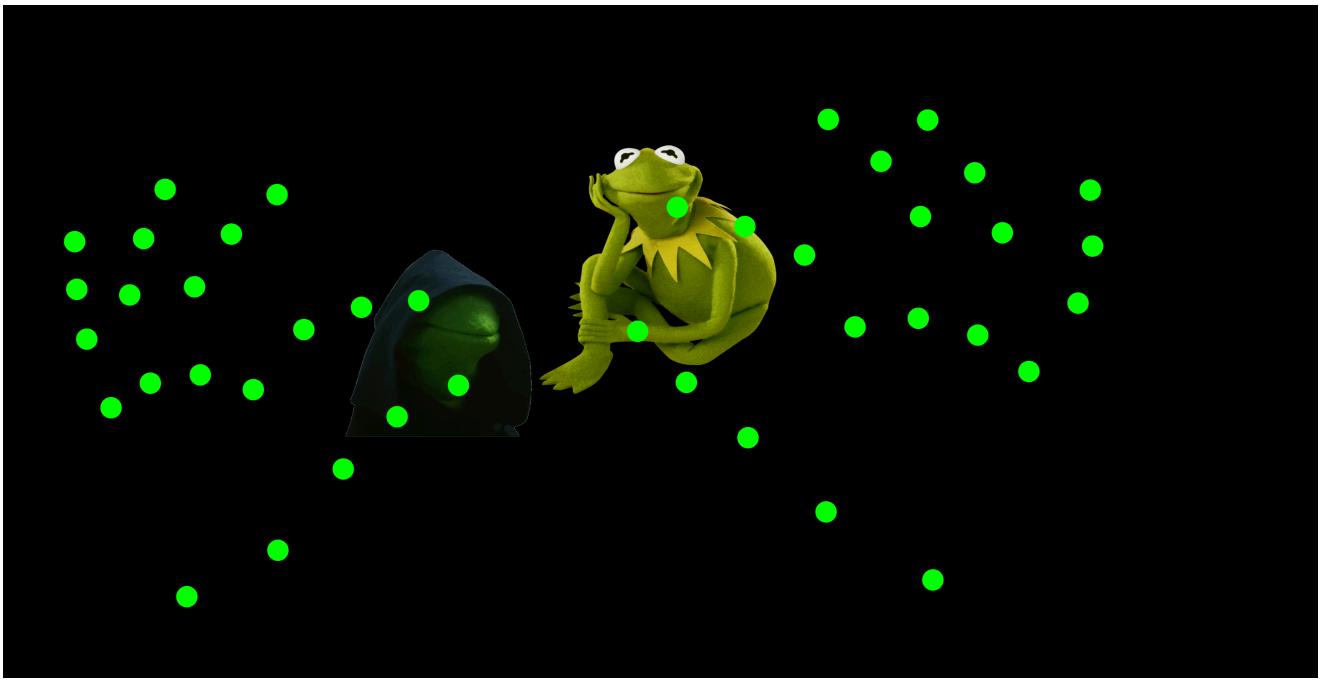

```
  translate(width, 0);
  scale(-1, 1);

  for (let i = 0; i < hands.length; i++) {

    let hand = hands[i];
 // fingertip points
    let thumb = hand.thumb_tip;
    let index = hand.index_finger_tip;
    // midpoint between fingers
    let x = (thumb.x + index.x) / 2;
    let y = (thumb.y + index.y) / 2;
// distance between fingers
    let d = dist(thumb.x, thumb.y, index.x, index.y);
    // image size reacts to pinch distance
    let size = d * 2;

    imageMode(CENTER);
    image(kermit, x, y, size, size);
  }
   // draw hand points
  for (let i = 0; i < hands.length; i++) {
    let hand = hands[i];

    for (let j = 0; j < hand.keypoints.length; j++) {
      let keypoint = hand.keypoints[j];

      fill(0, 255, 0);
      noStroke();
      circle(keypoint.x, keypoint.y, 30);
    }
  }
}

function gotHands(results) {
  hands = results;
}
```

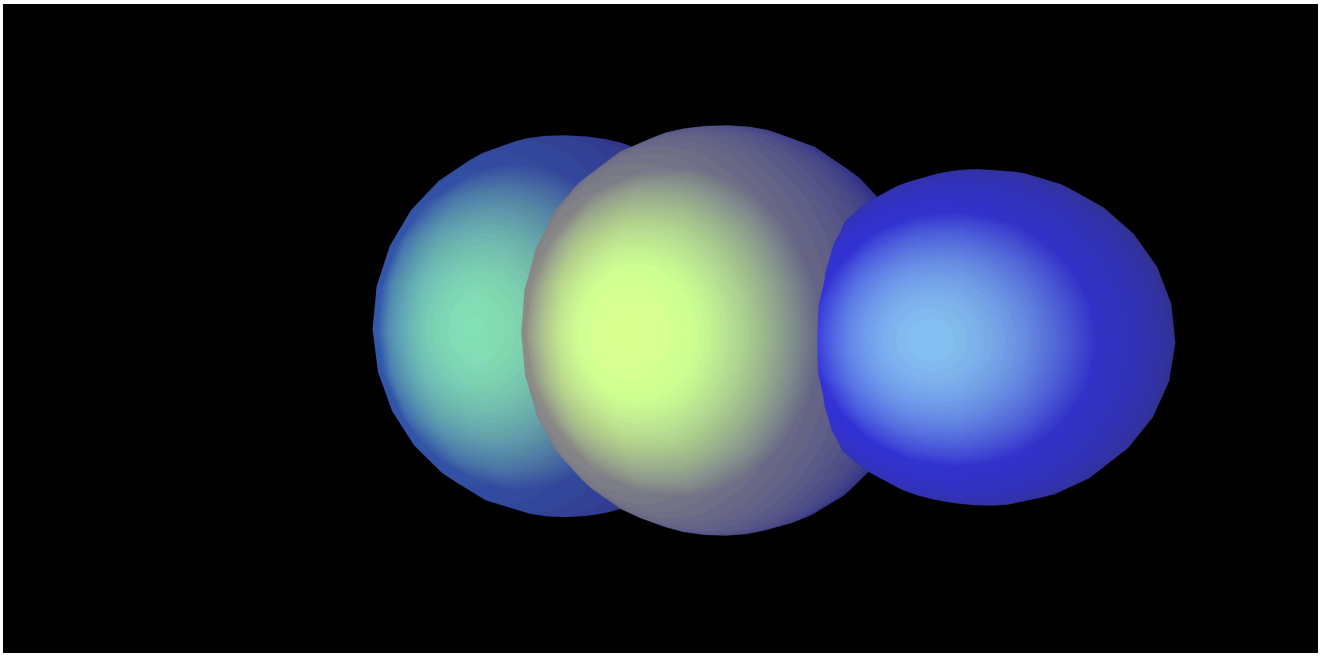


```
let libs = [
  'https://unpkg.com/ml5@1/dist/ml5.min.js',
  'https://unpkg.com/hydra-synth',
  'https://cdn.jsdelivr.net/gh/ffd8/hy5@main/hy5.js'
]

let handPose;
let video;
let hands = [];
let kermit;
let evilkermit;

function preload() {
  handPose = ml5.handPose();
  kermit = loadImage(
    'https://static.wikia.nocookie.net/alec-thaggard/images/e/e4/20388AA9-
245C-4E45-96B6-4C87EF411283.png/revision/latest?cb=20190617040210');
 evilkermit = loadImage('https://images-wixmp-
ed30a86b8c4ca887773594c2.wixmp.com/f/4a243a29-186e-46e2-b5fd-
3581c92b8930/dcpsjkv-7ceeda17-eea3-48a4-8aea-
a8a18d237cf7.png/v1/fill/w_413,h_420/png___the_muppets___evil_kermit_by_su
percaptainn_dcpsjkv-fullview.png?
token=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJ1cm46YXBwOjdlMGQxODg
5ODIyNjQzNzNhNWYwZDQxNWVhMGQyNmUwIiwiaXNzIjoidXJuOmFwcDo3ZTBkMTg4OTgyMjY0M
zczYTVmMGQ0MTVlYTBkMjZlMCIsIm9iaiI6W1t7ImhlaWdodCI6Ijw9NDIwIiwicGF0aCI6Ii9
mLzRhMjQzYTI5LTE4NmUtNDZlMi1iNWZkLTM1ODFjOTJiODkzMC9kY3Bzamt2LTdjZWVkYTE3L
WVlYTMtNDhhNC04YWVhLWE4YTE4ZDIzN2NmNy5wbmciLCJ3aWR0aCI6Ijw9NDEzIn1dXSwiYXV
kIjpbInVybjpzZXJ2aWNlOmltYWdlLm9wZXJhdGlvbnMiXX0.2gFJ7jyIYIWRhJQnQ0S2wXfz_
THfHWq3O0SGZadp0ww')

}
```

```
function setup() {
  createCanvas(windowWidth, windowHeight);

  video = createCapture(VIDEO);
  video.size(windowWidth, windowHeight);
  video.hide();

  handPose.detectStart(video, gotHands);
}

function draw() {
  background(0);

  translate(width, 0);
  scale(-1, 1);

 for (let i = 0; i < hands.length; i++) {

  let hand = hands[i];

  let thumb = hand.keypoints[4];
  let index = hand.keypoints[8];

  let x = (thumb.x + index.x) / 2;
  let y = (thumb.y + index.y) / 2;

  let d = dist(thumb.x, thumb.y, index.x, index.y);
  let size = d * 2;

  imageMode(CENTER);

  if (hand.handedness === "Left") {
    image(kermit, x, y, size, size);
  } else if (hand.handedness === "Right") {
    image(evilkermit, x, y, size, size);
  }
}

   // draw hand points
  for (let i = 0; i < hands.length; i++) {
    let hand = hands[i];

    for (let j = 0; j < hand.keypoints.length; j++) {
      let keypoint = hand.keypoints[j];

      fill(0, 255, 0);
      noStroke();
      circle(keypoint.x, keypoint.y, 30);
    }
  }
```

```
}

function gotHands(results) {
  hands = results;
}
```

3D, Lights (with Yann)



```
function setup() {
     createCanvas(windowWidth, windowHeight, WEBGL)

     // audio stuff now behind the scenes, 'true' makes class vars global
     setupAudio(true) // if empty, use 'a5.' before audio vars below
     // a5.ease = .075 // customize ease speed
}

function draw() {
     /* audio vars: amp, ampL, ampR, ampEase, fft, fftEase, waveform,
waveformEase */
     updateAudio()
     orbitControl)
     lights()
     ambientLight(10);
     ambientMaterial(100,100,255);
     noStroke()

     background(0)
     fill(0,0,255)


     push()
     specularMaterial(80,139,11)
```

```
     translate(200, 0)
     sphere(50 + fftEase[100])
     pop()


     push()
     fill(0, 100, 100)
     specularMaterial(80,139,11)
     translate(-100, 0)
     sphere(50 + fftEase[20])
     pop()

     specularMaterial(80,139,11)
     sph(0,fftEase[0] * .7, -50, 50 + fftEase[60])


     /* fftEase */
     for(let i = 0; i < fftEase.length; i++) {
          let freq = fftEase[i]; // (0, 255)
          let x = map(i, 0, fftEase.length, 0, width)
          let w = width / fftEase.length
          rect(x, height * .805, w, freq)
     }
}

function sph(x,y,z, size, rspeed){
     push()
     //rotate(frameCount * .1)
     noStroke()
     fill(255, 255, 0)
     translate(50, 0)
     sphere(50 + fftEase[60])
     pop()
}

/*
P5LIVE - Audio
If using outside P5LIVE, include p5live-audio.js
https://cdn.jsdelivr.net/gh/ffd8/P5LIVE/includes/utils/p5live-audio.js
*/
```